

BrickAtlas: Auditing Display-Grounded Evidence Following with Paired Observations of LDraw Parts

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Abstract

001 *Answering a visual question correctly does not establish*
002 *that a model tracks the evidence selected by its reference. A*
003 *paired test can address this gap only if the reference and the*
004 *answer remain well defined after the intervention. We intro-*
005 *duce a display-grounded observation contract on 24 human-*
006 *authored LDraw assemblies. Numeric labels select objects*
007 *in the current image; canonical source identities provide*
008 *audit provenance, not immutable answers to edited images.*
009 *Two separate families test rendered color changes at fixed*
010 *geometry (73 pairs) and candidate-reference reassignment*
011 *for geometry matching (67 pairs). A finite assignment opti-*
012 *mization removes a previously perfect option-order shortcut*
013 *and bounds 13 declared A-conditioned rules before infer-*
014 *ence. Human assignment prevents any reviewer seeing two*
015 *views of one parent. A frozen three-model roster binds ev-*
016 *ery planned run and failure to analysis. A supporting graph*
017 *audit demonstrates why the choice of control matters: an*
018 *isolates-plus-one heuristic solves 72/73 natural questions*
019 *and all 61 degree-visible change pairs, but has zero both-*
020 *correct on 12 degree-matched pairs; BFS solves all of them.*
021 *The strong subset is a small proof of concept, including an*
022 *anonymous 1-WL blind spot. We release machine-verified*
023 *stimuli, the executable evaluation protocol, and a search-*
024 *able 617-item historical dossier. Human visual certifica-*
025 *tion and learned-model diagnostic value remain empirical*
026 *requirements; the reported findings concern stimulus and*
027 *algorithmic audits.*

028 1. Introduction

029 A label may identify a source instance or select a visible ob-
030 ject. These meanings diverge under intervention: relabeling
031 another part B0035 does not change the original B0035's
032 type. Changing gold without specifying the referent risks
033 rewarding interpretation shifts.

034 We ask a narrower, observable question: *does a response*
035 *correctly follow the property or candidate reference se-*

lected in the current display, when that observation changes 036
under an explicit contract? 037 The contract resolves numeric
038 references within the supplied image. The source identity
039 is retained for construction, pairing and inspection. The
040 model is never asked to recover a historical source fact from
041 a deliberately edited display.

042 We study two atomic visual families separately (Fig-
043 ure 1). In Color, the same original geometry and pose are
044 rendered with two opaque body colors, while the camera, la-
045 bels and material settings are fixed. The question explicitly
046 concerns the current rendered appearance. In Part-type, tar-
047 get geometry and its label are fixed, while two candidate la-
048 bels exchange their visible referents. The correct candidate
049 token changes; no canonical part changes its type. These in-
050 terventions support different interpretations, so they never
051 contribute to a pooled visual score.

052 Paired image-dependent answers have substantial prece-
053 dent in VQA [3], contrast sets [2], and joint grounding
054 tests [12]. Our contribution is not the invention of pairing
055 or both-correct scoring. It is a versioned measurement con-
056 tract that connects a well-defined current referent, condition-
057 specific gold, independent review of both exact observa-
058 tions, and receipt-derived response transitions in a reusable
059 assembly resource. This connection matters because correct
060 gold in a CAD record does not guarantee a uniquely visible
061 answer.

062 The supporting graph audit gives a concrete example of
063 how inadequate controls can defeat the intended interpreta-
064 tion. A shallow rule that counts isolated surviving nodes
065 and adds one for any remaining nodes answers 72/73 natural
066 component questions correctly. It also solves both sides of
067 all 61 change pairs whose degree distributions differ. On 12
068 pairs matched in complete degree sequence, the same rule
069 has zero paired both-correct, while component algorithms
070 solve all pairs. This establishes an interpretive distinction
071 between the distributions. It does not establish that a learned
072 model uses the heuristic.

073 Pairing alone is insufficient. An earlier Part-type assign-
074 ment always chose the first candidate different from A's
075 gold, permitting perfect Both without reading B. This revi-

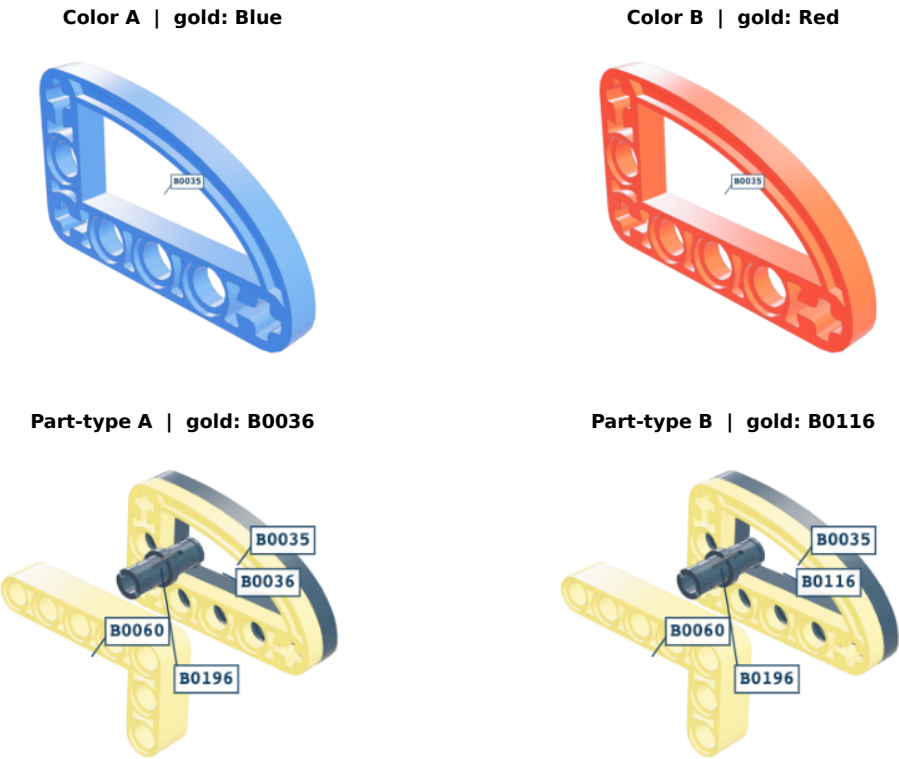


Figure 1. Actual paired observations from OMR 42004, shown with identical within-family paper crops. Top: Color asks for the current color of B0035; only the rendered body-color parameter changes, from Blue to Red. The original source color remains Black in provenance. Bottom: Part-type keeps target B0035 fixed and swaps candidate references B0036 and B0116. The matching geometry is unchanged; the answer changes from B0036 to B0116. Bottom details omit the remote swapped candidate. Gold annotations and crops belong only to this figure. Model inputs are the full 1280×800 PNGs and new display-grounded questions. These are stimulus examples, not model response cases or human answerability judgments.

sion jointly balances choice transitions, audits explicit short-cut predictors, and freezes model inclusion before inference. Its contributions connect display semantics, bounded stimulus design and complete evidence collection. Whether this protocol reveals learned-model differences beyond ordinary accuracy remains empirical.

2. Related Work and Measurement Scope

Assembly benchmarks. BrickNet [6] represents placed brick instances and connectors in graph-backed generative assembly programs. LEGO-Puzzles [11] evaluates spatial/sequential reasoning in editable scenes with three-annotator validation in its pinned v3 protocol. PhyBlock [8] studies physical planning with dependencies and simulation. Numeric IDs and reusable scene assets are established machinery; our endpoint concerns two elementary visual properties, not assembly planning.

Contrastive and consistent answers. Contrast sets probe local decision boundaries using small meaningful input

changes [2]. VQA v2 pairs similar images that support different answers to the same question, reducing the utility of language-only shortcuts [3]. Cycle-consistency and VQA-Rephrasings address answer stability under linguistic variation [10]. Required answer change and required invariance are different tests: a constant predictor can be stable without responding to evidence. BrickAtlas records which property an intervention changes and which answer policy applies, rather than treating every response change as either success or failure.

Reference grounding and construct validity. Numeric labels are narrower than referring-expression comprehension [9] or Winoground’s joint image–caption matching [12]. Small objects, resolution and off-construct requirements can explain grounding failures [1]; independent review therefore judges exact questions and pixels, including visual answer uniqueness.

Controlled visual and graph evidence. CLEVR [5] and GQA [4] illustrate the value of explicit scene structure and question operations. BrickAtlas uses source structure for provenance while keeping it out of visual requests. A separate text-graph challenge tests component counting. The expressivity analysis of neighborhood aggregation against Weisfeiler–Lehman refinement [13] clarifies a limit of that challenge: our six-cycle/two-triangles pairs are indistinguishable to anonymous uniform-feature 1-WL. This is a known representational limitation instantiated by the audit, not a new impossibility result or an expectation that a simple GNN must solve the pairs.

3. Resource and Reference Contract

3.1. Preserved sources and separate roles

The historical resource has 617 questions on 24 human-authored assemblies and 15,334 placed parts in the LDraw format [7]. Original files, source transforms, questions and answers remain unchanged. The 140 visual parents define observations in display-v3; the other 477 historical items remain accessible controls and structural tasks. A historical task ID is a provenance key, not a key for transferring a response into the new contract.

Visual questions receive image/text, structural questions explicit graphs. Historical record-reading and visibility controls remain separate. Shared IDs connect these artifacts without making each question a perception-to-action chain.

Source bands D1–D4 describe assembly scale, not cognitive difficulty. The supplement lists every source; the separate dossier lists all families. The stable schema key `shape-match` displays as Part-type.

3.2. Four meanings that must remain distinct

For a source hash h and instance ID i , canonical identity $c = (h, i)$ identifies the original record. Source truth $T(c)$ stores its type, color and transform. In observation o , a display mapping $d_o(\ell)$ selects the visible referent of label ℓ . The scored answer y_o is a function of the question and this observation, not necessarily of $T(c)$ alone.

Every new visual question includes the following rule: *References select objects in the supplied image only. A label such as B0001 is an image-local display reference, not a request to retrieve a historical source instance. Renders may have edited appearances or label assignments. Answer only from the current image.* This rule is part of the hashed model-visible text on both sides. The evaluator retains c , $T(c)$ and the display mapping outside the request.

Color asks, “What color is the object currently marked B0035 in this image?” Part-type asks, “Which displayed candidate label currently marks an object with the same visible part geometry as the target marked B0035? Ignore color and pose.” The complete Part-type prompt also repeats that

the current image, rather than historical identities, determines the answer. Options have explicit choice IDs. The response contract requires exactly one JSON object containing a listed `choiceId`.

3.3. What is physically preserved

Source geometry and poses are immutable. Isolated or operand-only views may omit supports; they are observations, not freestanding configurations. Color edits and separate part panels propose no new physical assembly.

The supplied connector graph covers recognized relations, not every contact or stability condition. The historical inventory still contains 111 other flagged tasks and 573 intersection candidates requiring human decisions. A candidate may be intended mating, an approximate mesh intersection or a source defect. This revision does not resolve those physical judgments through visual benchmark code.

4. Interventions and Exact Observations

4.1. Color: controlled rendered appearance

The 73 Color pairs use the same original geometry, source pose, camera, scale, label projections, visibility and material settings. Only the body-color parameter changes between two distinct named opaque colors. Both arms explicitly permit edited appearance; neither arm asks for the original source color. The 146 observations use Red, Blue, Yellow and Black, with aggregate counts 37, 37, 36 and 36 respectively. These are displayed gold categories, not a sample of natural material colors.

Color retains its display-v2 image/text bytes. It tests named rendered appearance, not illumination invariance, material recognition or recovery of natural object color.

The machine audit checks common camera and label metadata and compares the two native images. All 73 pairs have zero changed pixels outside the common projected body mask dilated by one antialiasing pixel. Independent regeneration of three canary observations is byte-identical. These checks locate the pixel change; they cannot establish that humans perceive every chosen category unambiguously.

4.2. Part-type: candidate-reference reassignment

The 67 Part-type pairs retain the target geometry and token. The matching candidate exchanges labels with one other candidate, so the current reference that selects matching geometry changes. The option list remains fixed within the pair. Source part numbers independently derive a candidate gold, but the question asks for visible geometry. Similar types, hidden features or a confusing label leader can defeat visual uniqueness even when the record-level match is exact.

Table 1. Choice-ID gold counts beside the planned experiment inventory. Color is frozen; only Part-type was optimized. Both families cover 24 sources. Actual human qualification is 0/280 primary endpoints.

Family	Arm	A	B	C	D
Color	A	13	22	19	19
Color	B	16	15	20	22
Part-type	A	16	17	17	17
Part-type	B	17	17	17	16

Both Part-type arms receive new option orders, image renders, observation IDs and review lineage. Base geometry pixels match exactly within each pair; changed pixels lie within the label masks. Two regenerated canaries are byte-identical. In Figure 1, B0035 stays fixed while the matching candidate’s label changes from B0036 to B0116. No old Part-type review or response is transferable.

4.3. Joint assignment and bounded shortcut audit

For each parent we enumerate all four-option permutations and all nonmatching swap candidates already visible in the operand set, with an on-screen anchor and exactly one source-type match. The 4,824 configurations reduce to 2,412 constraint-equivalent representatives. Binary selection chooses one configuration per parent. Bounds are committed before optimization: each arm has 16 or 17 golds per position, each off-diagonal transition has 5 or 6 pairs, and diagonal counts are zero. Every fixed-choice predictor must score at most .30 source-macro on either arm. Thirteen declared oracle-A rules must have B accuracy at most .40 in both micro and source-macro estimates.

The objective minimizes summed integer SHA-256 configuration costs; sorted inputs and the checked single-thread HiGHS runtime give the recorded optimum 1,007,668. The immutable assignment is authoritative across runtimes. Rules use option order, alternate label numbers or nearest/farthest A-side source centers and projected anchors; none reads B pixels. Independent gold/feature reconstruction enforces the contract before sealing, live inference and publication. These are design bounds, not significance thresholds or evidence against every possible shortcut.

Tables 1 and 2 expose actual priors. The former perfect first-alternative attack now has 23/67 Both (34.33% micro; 26.94% source). Because each transition cell is at most 6, any single deterministic mapping from A choice alone to B choice has in-sample accuracy at most $4(6)/67 = 35.82\%$. This restricted ceiling does not apply to predictors using labels, source IDs or additional features. Color was not reoptimized: its last-alternative and cyclic previous rules reach 32/73. Balanced semantic color totals therefore do not imply balanced choice transitions or a shortcut-free family.

A separate set of 67 position-reference observations

Table 2. Deterministic B accuracy (%), micro and equal-source.

[†]Oracle supplies correct A; these rules read no B image, so their B accuracy equals Both. Fixed choices have Both=0. These are stimulus audits, not learned-model scores. All 13 Part-type oracle rules, intervals and individual errors are released.

B predictor	Color		Part-type	
	Micro	Source	Micro	Source
Fixed A	21.92	18.47	25.37	20.28
Fixed B	20.55	20.76	25.37	27.71
Fixed C	27.40	31.39	25.37	27.50
Fixed D	30.14	29.38	23.88	24.51
First alternative [†]	26.03	23.47	34.33	26.94
Last alternative [†]	43.84	43.12	32.84	32.01
Cyclic next [†]	30.14	31.46	35.82	38.68
Cyclic previous [†]	43.84	43.61	34.33	28.89
Cyclic two [†]	26.03	24.93	29.85	32.43

shows a target and four candidate panels without text inside the image. Options refer to candidate positions. This provides a geometry-matching comparison under a different reference interface. Panel layout and visual access also change; a difference from label-based performance cannot by itself identify an OCR or label-tracking effect. The control has its own human reviews and never enters primary paired Both.

4.4. Freezing the input boundary

An exact observation includes the complete system/user messages, question, options, native 1280×800 PNG bytes, resolution and image-detail policy. All 347 observations are frozen before review: 280 primary endpoints and 67 auxiliary controls. Rendering metadata, canonical source truth, conditions, pairing and gold are evaluator-only records. Neither endpoint is supplied as conversational history to the other.

The reviewed observation is distinct from the provider transport envelope. A run pins its model revision, endpoint adapter and generation settings, and separately hashes the full HTTP request. The runner must reproduce the reviewed messages and image bytes inside that envelope. This supports comparisons across pinned models without asserting that a provider’s internal image preprocessing is observable. Any change to reviewed text, pixels, dimensions or image policy requires new review; changing only the pinned provider envelope does not create a new visual question.

5. Human Qualification and Response Analysis

5.1. Independent review of both sides

Six distinct reviewers receive 115, 116, 114, 115, 115 and 119 units, respectively, in shuffled queues with opaque unit IDs. Every observation receives two judgments, but nobody

sees more than one observation of a parent, including the auxiliary panel. Thus A/B reviewer sets have empty intersection; reviewer slots do not identify an arm. The interface shows one observation at a time, its exact text and scrollable native pixels. It withholds the condition name, parent/pair membership, source record, gold, model output and the other reviewer’s decision. The primary denominator is always 280; it cannot become a B-only queue through task-ID deduplication.

A raw review contains the reviewer pseudonym, timestamp, categorical decision, visual choice, reason, and attestations of native-size inspection and independence. Four structured checks ask whether the target reference is unique, exactly one option is visually correct, visibility is sufficient, and the observation lacks reference contradictions. Decisions distinguish decidable, ambiguous, unanswerable and contradictory cases.

Matching decisions, choices and checks produce a final judgment. Any disagreement among these fields requires explicit adjudication. Up to three additional people occupy adjudicator slots; none is an initial reviewer or sees multiple observations of a parent. Assignment, imports and final reconstruction all verify this exposure rule. Both original records and raw adjudication exports remain intact. A final decidable choice that conflicts with the generated gold blocks finalization and requires revising the item and repeating QA, rather than silently replacing the answer. Agreement reports include raw decision agreement, multiclass Cohen’s kappa, choice/check agreement and their actual denominators. Undefined kappa remains undefined.

The runner requires final decisions for every planned endpoint before real visual inference. Nondecidable final judgments remain in the full denominator; only pairs with two final decidable endpoints enter the additional QA subset. This prevents post-result item selection. The gate recomputes decisions from retained raw files. It verifies recorded provenance and attestations, not a person’s identity through cryptography. Coordinator oversight remains necessary.

5.2. Correct adaptation and failure denominators

For pair i , let p_{Ai}, p_{Bi} be normalized choices, $y_{Ai} \neq y_{Bi}$ their golds, and v_{Ai}, v_{Bi} indicate valid output. Invalid or missing predictions are never equal to a gold. For N fixed pairs, we report

$$A\text{Acc} = N^{-1} \sum_i [p_{Ai} = y_{Ai}], \quad (1)$$

$$\text{NewAcc} = N^{-1} \sum_i [p_{Bi} = y_{Bi}], \quad (2)$$

$$\text{Both} = N^{-1} \sum_i [p_{Ai} = y_{Ai}][p_{Bi} = y_{Bi}]. \quad (3)$$

The B outcomes partition into new-gold correctness, old-gold retention, another valid choice and invalid output. In particular,

$$R_{\text{old}} = N^{-1} \sum_i v_{Bi} [p_{Bi} = y_{Ai}]. \quad (4)$$

Old-gold retention is a behavioral outcome, not proof of memory or an internal causal mechanism.

Conditional adaptation is Both divided by A accuracy for the micro estimate. Its denominator is the number of A-correct pairs; a zero denominator yields null. Conditional source-macro analysis averages within-source conditional rates only over sources with eligible pairs and reports that coverage. We retain unconditional estimates so conditioning cannot hide a weak A endpoint.

Format-valid rates are reported for A, B and both endpoints, alongside semantic rates. Timeouts, API failures, refusals, missing receipts and illegal formats remain incorrect in the planned denominator, with separate failure reasons. A stable invalid string is not semantic consistency. The auxiliary position control reports its own accuracy and validity without being pooled with either visual family.

5.3. Aggregation, selection and comparisons

For a pair outcome z_{si} from source s , the primary estimator is

$$\hat{\mu} = \frac{1}{|S|} \sum_{s \in S} \frac{1}{n_s} \sum_{i=1}^{n_s} z_{si}. \quad (5)$$

Each family has one primary pair per parent. We also report micro rates, pair counts and source counts. A family absent from a source does not contribute an artificial zero. Percentile intervals use 10,000 whole-source bootstrap draws with seed 20260920. Resampling sources preserves within-source dependence. We also freeze 18 primary author groups and 17 contributor-connected groups from the source credits. Sensitivity reports include equal-author means, author-cluster intervals for both weighting schemes, and leave-one-author-out differences. Contributor clustering also includes OMR editors. All 276 source-pair part-inventory Jaccards are published; inventory overlap is not aligned geometry. A direction changing under author omission limits any cross-source stability claim.

Full and both-decidable estimates are published together with exclusion reasons and source coverage. The QA-minus-full difference resamples the same source weights for both estimates, recomputing each mean on its eligible sources. This retains changes in source composition. Overlapping exclusion reasons are not added as independent counts. Empty subsets and bootstrap draws without selected sources are explicitly recorded.

Model comparisons use paired outcomes on the same items. Ordinary-A, NewAcc and Both ranks use exact rational point comparisons and descending midranks for ties.

Table 3. Frozen complete roster. OpenAI is the provider for every row. Runs means completed/planned; each plan includes visual, no-image and text-graph conditions. No collection dates or scores exist yet. The complete inclusion and failure policies are frozen in the roster.

Model	Pinned revision	Runs
GPT-4.1	gpt-4.1-2025-04-14	0/3
GPT-4o	gpt-4o-2024-08-06	0/3
GPT-4o mini	gpt-4o-mini-2024-07-18	0/3

A reversal means the signs of the A and Both differences disagree; a tie is separately classified. Paired source intervals accompany the differences. We also record whether comparison directions agree between full and QA subsets. The intervals are descriptive and unadjusted for multiple comparisons. An interval containing zero does not support a directional ranking. The estimand is one frozen invocation set, not cross-call stability; temperature zero does not establish provider determinism.

5.4. Frozen roster and image-dependency control

Table 3 specifies three OpenAI snapshots, chosen to compare an instruction-following reference, an earlier multimodal model and its smaller tier under one implemented transport. This is not a current-model or cross-provider leaderboard. Each has one 347-observation visual run, one 347-observation no-image control and one 219-observation graph run: exactly nine runs and 2,739 scheduled calls. The no-image condition removes only image content, retaining question/options. It measures dependency on supplied images, not independently answerable visual tasks.

All adapters pin Chat Completions, exact requested/returned revisions, high-detail native images, temperature zero, 128 output tokens, a 90-second timeout and one attempt with no retries. Official model documentation was checked; live account access remains untested. Analysis requires exact equality of observed and planned run sets. Every entry binds adapters, requests, raw receipts, QA and report rows through a model-run index. Missing or extra runs block empirical tables. Interrupted runs require explicit closure, retaining pre-call intents and missing-attempt records in the original denominator. They cannot be replaced by a favorable repeat.

5.5. What would establish diagnostic value?

A useful result would reveal differences obscured by ordinary accuracy, robust to answer priors, image withdrawal, QA selection and author dependence. If Both merely reproduces A rankings or differences reflect format/API failures, added diagnostic value is limited. Deterministic case extraction links the first hash-ordered QA-decidable example of each response transition to its requests and raw receipts;

Table 4. Current primary visual inventory. QA final counts completed human decisions, not rendered images. Completed model runs are 0/9 planned. Both-decidable coverage is 0/140 pairs; this is an evidence-availability count, not model accuracy.

Family	Pairs	Sources	QA final
Color	73	24	0/146
Part-type	67	24	0/134

absent types remain absent. No learned rankings, reversals or response cases exist yet. Table 4 records evidence availability.

6. Graph Audit: Which Controls Resist Shallow Rules?

6.1. Natural evidence and hypothetical partitions

A historical Graph deletion question asks for the number of connected components after deleting a specified node from an explicit undirected graph, including isolated survivors. Duplicate edges are deduplicated. Independent BFS and union-find implementations agree on all golds. These algorithms validate the supplied graph contract, not the completeness of the physical connector detector.

The audit retains all 73 natural questions from 24 sources. It also reclassifies the existing 219 hypothetical observations: each parent has an anchor, an answer-changing partner and an isomorphic answer-preserving partner. The 12 strong change pairs span eight sources and match surviving node set, edge count, target degree, complete surviving degree multiset and isolate count. The other 61 change pairs span 23 sources and are explicitly called *degree-visible controls*. Their invariance pairs are reported separately. The shared anchors mean these four partitions cannot be summed as independent observations.

Strong pairs compare a six-cycle with two triangles, with identical extra isolated nodes. Weak pairs use a path versus a triangle plus isolates. Both templates can match node and edge counts, but only the former matches degrees. These are hypothetical edge lists derived for a controlled test; they are not claims about newly discovered physical connections in the source assemblies.

6.2. Baselines and matching limits

We include node-count, edge-count, degree-histogram and anonymous 1-WL feature lookups; the isolates-plus-one rule; and BFS/union-find. For a surviving graph G' , the heuristic is

$$\hat{c}_{\text{iso}} = n_0 + \mathbf{1}[|V(G')| > n_0], \quad (6)$$

where n_0 is its isolate count. It does not traverse components. Feature lookups use labels only from other sources,

Table 5. Computed algorithm outputs, not learned-model results. Natural correct items and hypothetical both-correct pairs use distinct source-held-out fits. Strong change is a small proof of concept; degree-visible controls support a weaker claim. Invariance reuses anchors.

Algorithm	Natural correct / 73	Strong change Both / 12	Degree-visible Both / 61	Strong invariance Both / 12
Node-count lookup	63	0	0	6
Edge-count lookup	44	0	0	7
Degree-histogram lookup	69	0	61	6
Anonymous 1-WL lookup	69	0	61	6
Isolates + one	72	0	61	6
BFS	73	12	61	12
Union-find	73	12	61	12

select the most frequent answer, break ties toward the smaller integer, and abstain on unseen features. Abstentions count as incorrect. Training uses changed endpoints once; invariance copies do not duplicate training labels. Natural and hypothetical fits remain separate.

Anonymous uniform-feature 1-WL gives identical signatures on all strong change pairs. A deterministic predictor restricted to that signature must give the same answer to different golds, hence Both=0. This also limits message passing no more expressive than 1-WL [13]. Unique IDs, positions, spectral and higher-order methods are outside the claim; no trained GNN result is implied.

6.3. Measured results and uncertainty

Table 5 shows that isolates-plus-one obtains 72/73 natural correct answers, compared with 73/73 for BFS. Its natural source-macro accuracy is 97.92%, with a descriptive 95% source interval [93.75, 100]. Degree-histogram and 1-WL lookups reach 69/73. Natural accuracy barely distinguishes traversal from these shallow rules.

On degree-visible change pairs, degree-histogram lookup, 1-WL lookup and isolates-plus-one all achieve 61/61 Both. On strong change pairs, each obtains 0/12; BFS and union-find obtain 12/12. Strong invariance illustrates another distinction: the degree and 1-WL predictors are consistent across the two isomorphic observations, but only 6/12 pairs are jointly correct. Invariance of predictions alone would reward the six stable wrong pairs.

Strong change directions have six increases and six decreases, although raw arm histograms differ. Unordered Both and symmetric endpoint accuracy retain the 12-pair denominator. Histograms, all strong pairs and invariance results appear in the supplement.

Approximate pair Wilson intervals are [0, .2425] for 0/12 and [.7575, 1] for 12/12; pair independence is imperfect. Whole-source intervals can degenerate at boundary outcomes. Neither establishes generality beyond eight sources and one template.

7. Reproducibility, Limits and Empirical Acceptance

Earlier release bytes and source render assets are archived and hashed. New Part-type IDs prohibit transferring earlier reviews/responses. Empirical analysis reparses raw outputs, validates served revisions and checks the full roster, gold provenance and common QA snapshot. Dry runs and fixtures cannot enter empirical reporting.

Machine verification covers 347 frozen visual observations, independently derived gold and shortcut rules, label/body pixel audits, all 2,739 planned request envelopes and 27 new protocol/analysis tests. The 26 historical tests remain preserved. These checks establish implementation behavior, not real human agreement or remote-provider authenticity. Collection oversight and retained provider records must support receipt provenance.

The release has zero genuine human reviews and 0/9 completed model runs. The primary empirical question remains unresolved. Actual judgments, full/QA results, retention, format effects and genuine cases are required.

The supplement gives protocol details, source coverage and all strong pairs. The separate 617-item dossier links to a 3D viewer with rotation, zoom, expansion and action replay. Those inspection views do not substitute for native-input human qualification.

The bounded scope is instance-label-grounded evidence following on two atomic visual attributes and 24 sources. Rendered colors do not cover natural color constancy; candidate-label changes do not establish general referring-expression understanding. The auxiliary panel comparison changes visual access. Graph evidence concerns a limited recognized graph and a small hypothetical challenge. Public source reuse also leaves contamination and cross-source dependence concerns. These limits bound what future positive results could establish.

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